Advanced Quantitative Text Analysis

Day 1: From Text to Measurement

Petro Tolochko

Summer School Structure

1. **Day 1:** Text representation, preprocessing, feature engineering
2. **Day 2:** Dictionaries, annotation, and supervised classification
3. **Day 3:** Unsupervised learning: clustering, topic models, scaling
4. **Day 4:** Embeddings, transformers, multilingual analysis, LLMs
5. **Day 5:** Evaluation, fine-tuning, causality, research design

Course Structure

- Theoretical input
- Practical sessions
- Discussion of participants' own projects
- Focus on method choice, validation, and research design

Introduction

- Post-Doc, Computational Communication Science Lab, University of Vienna
- Research focus: text complexity, social networks, statistical modeling
- petro.tolochko@univie.ac.at

Introduction: Participants

- Name?
- Disciplinary background?
- Experience with text analysis?
- Programming experience?
- Expectations / wishes for the summer school?

Objectives

- Understand major approaches to automated text analysis
- Learn how text is transformed into data
- Identify practical challenges in real text-analysis projects
- Critically evaluate representation and modeling choices
- Connect methods to your own research questions

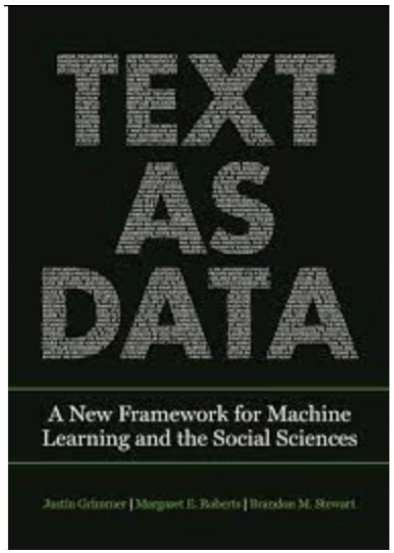
Course Resources

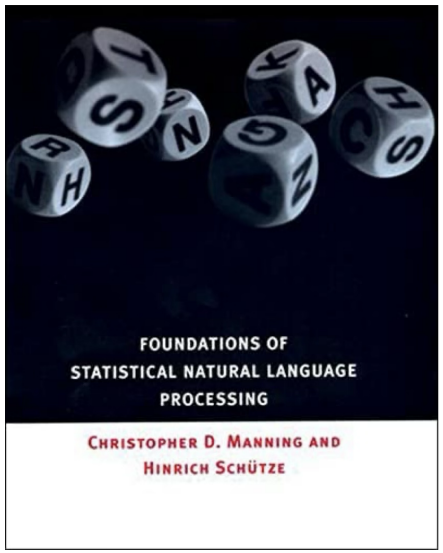
- Course repository:
- `https://github.com/PeterTolochko/MethodsNET_Summer_School`
- Code, datasets, notebooks, and exercises will be organized there.

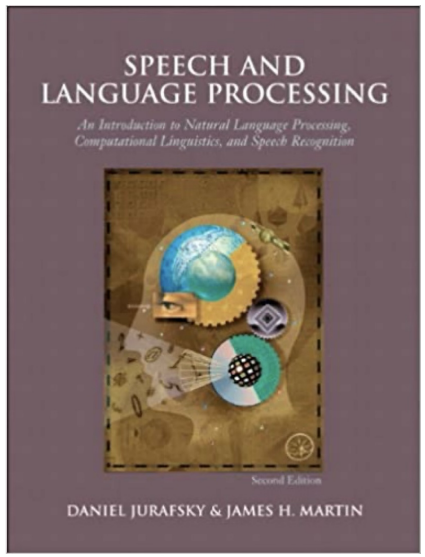
Today

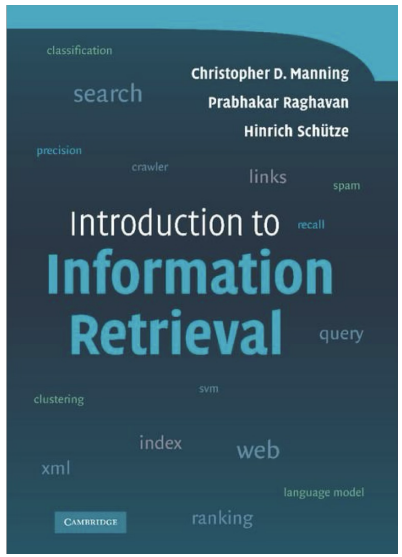
- Text as data, text as measurement
- Document-term matrices and bag-of-words
- Preprocessing as an analytical choice
- TF-IDF and feature weighting
- Similarity, distance, and comparing texts
- Practical: how representation changes results

Resources

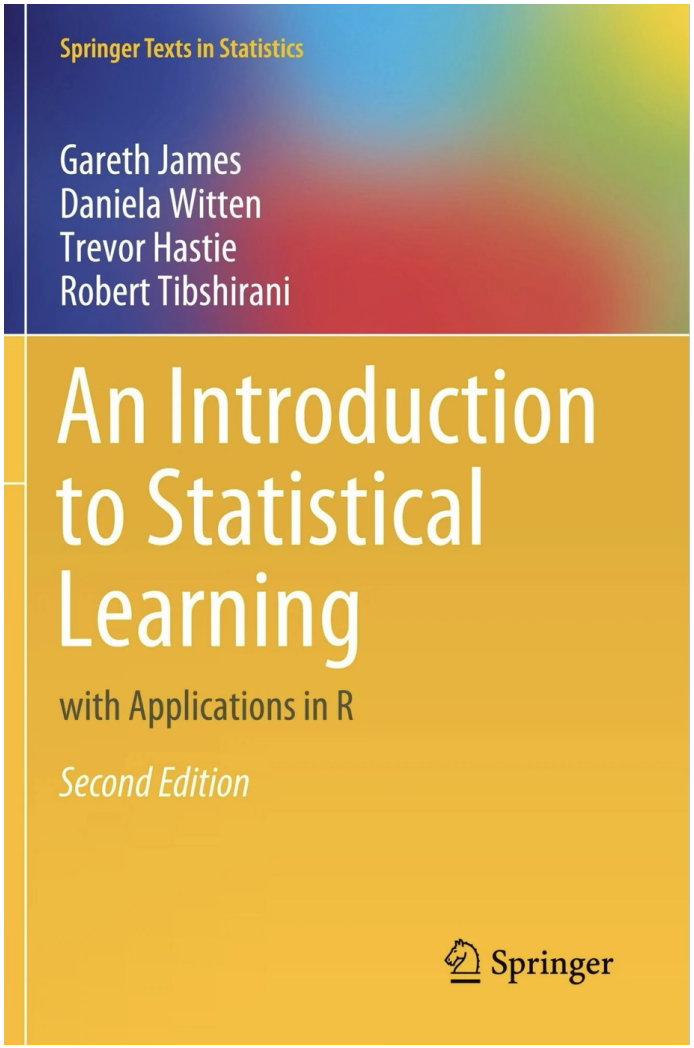

TEXT AS DATA
A New Framework for Machine Learning and the Social Sciences
Justin Grimmer | Margaret E. Roberts | Brandon M. Stewart

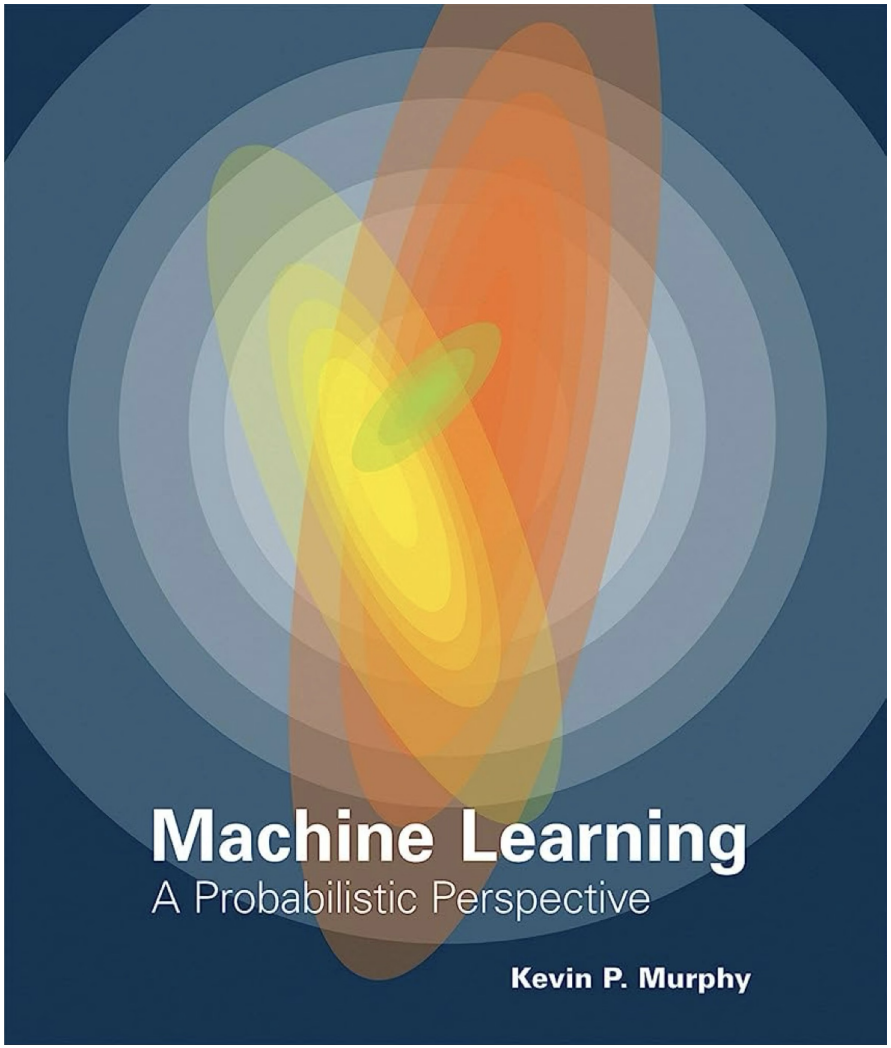

FOUNDATIONS OF STATISTICAL NATURAL LANGUAGE PROCESSING
CHRISTOPHER D. MANNING AND HINRICH SCHÜTZE


SPEECH AND LANGUAGE PROCESSING
An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition
Second Edition
DANIEL JURAFSKY & JAMES H. MARTIN


classification
search
Christopher D. Manning
Prabhakar Raghavan
Hinrich Schütze
precision
crawler
links
spam
Introduction to
recall
Information Retrieval
query
clustering
svm
index
web
xml
language model
CAMBRIDGE
ranking

Resources


Springer Texts in Statistics
Gareth James
Daniela Witten
Trevor Hastie
Robert Tibshirani
An Introduction to Statistical Learning
with Applications in R
Second Edition
Springer


Machine Learning
A Probabilistic Perspective
Kevin P. Murphy

Resources

- *Advanced Data Analysis from an Elementary Point of View*, Cosma Rohilla Shalizi (free online)
- `https://www.stat.cmu.edu/~cshalizi/ADAfaEPoV/`

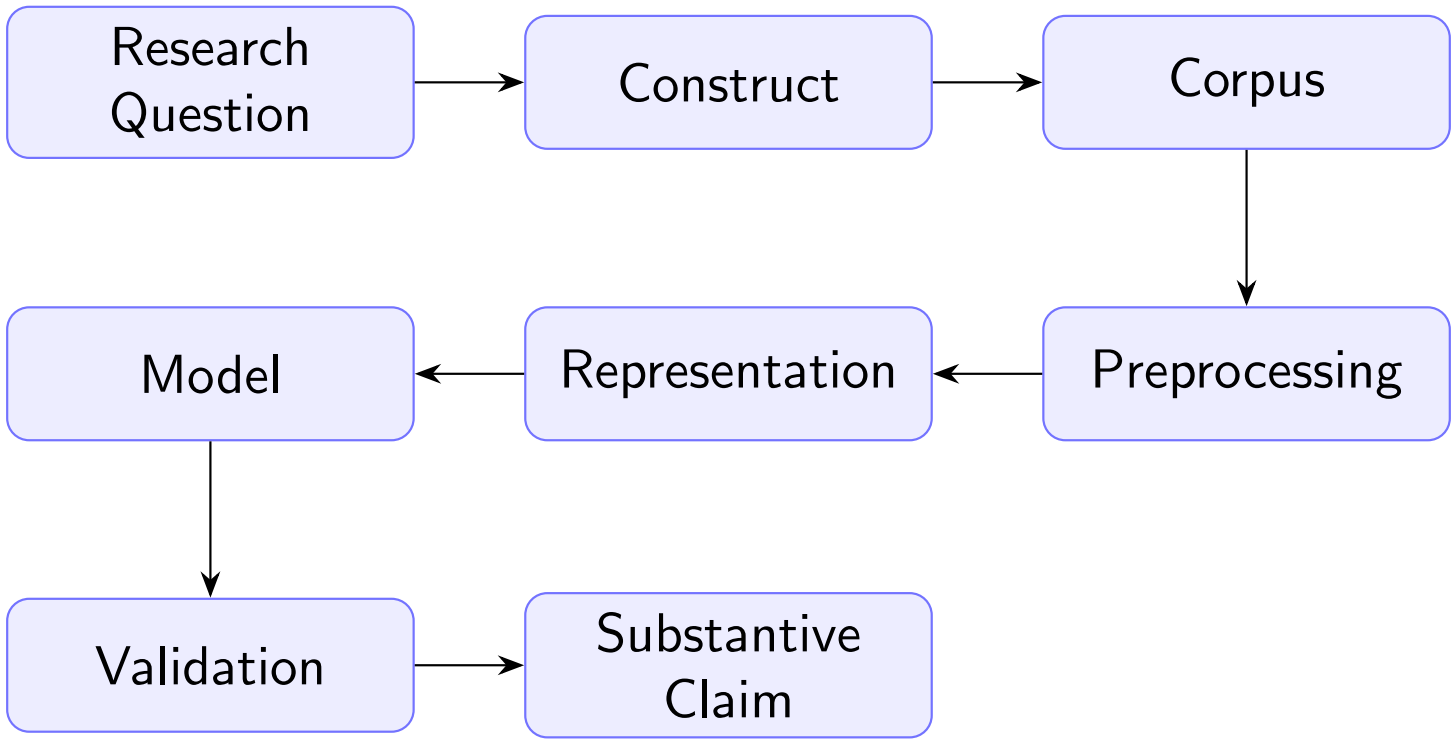
Why Analyse Texts?

- Vast quantities of textual data are available
- Texts carry meaning
- Texts capture social, political, and cultural behaviour
- Texts can help us study phenomena that are otherwise difficult to observe

What Do We Measure Through Text?

- We usually do not analyse text "for its own sake"
- We use text to measure latent or difficult-to-observe concepts:
  - attention sentiment framing ideology
  - complexity toxicity uncertainty polarization
- The representation determines what the model can measure

The Text Analysis Pipeline



The Basic Problem

- Text is meaningful to humans
- But computers do not directly "understand" text
- We need to transform text into structured representations
- Different research questions require different representations

What is Text?

- Data
- Unstructured
- Highly multidimensional
- Difficult to work with if you are not human

From Text to Structure

- We need to structure text before analysis
- We need representations that computers can process
- We also need representations that remain meaningful to us
- Different research questions $\rightarrow$ different representations

Unit of Analysis

Research question	**Possible unit**
Does newspaper coverage become more negative?	article / paragraph / sentence
Do parties use simpler language?	speech / manifesto / paragraph
Are social-media posts toxic?	post / comment
What topics dominate climate discourse?	article / document / sentence

- Unit choice determines what the model can see

Document-Term Matrix

$$X = N \times K$$

- N = number of documents
- K = number of terms/features
- X = structured representation of the corpus

Document-Term Matrix

$$X = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$N = 3, \qquad K = 6$$

Example Corpus

- Doc 1: "John loves ice-cream"
- Doc 2: "John loves oranges"
- Doc 3: "Mary hates ice-cream"

$$N = ?, \qquad K = ?, \qquad X = ?$$

Document-Term Matrix with Vocabulary

	John	loves	ice-cream	oranges	Mary	hates
D_1	1	1	1	0	0	0
D_2	1	1	0	1	0	0
D_3	0	0	1	0	1	1

$$N = 3, \qquad K = 6, \qquad X \in \mathbb{R}^{3 \times 6}$$

Types & Tokens

- **Tokens:** all word occurrences
- **Types:** unique word forms

Example corpus:

- 9 tokens
- 6 types

Bag-of-Words Representation

- Representation of text as a bag of words
- Word order is ignored
- Each document is represented as counts of words
- Simple, powerful, and still widely used

What Bag-of-Words Captures and Loses

Captures

- word frequency
- topical emphasis
- repeated lexical choices
- broad document similarity

Loses

- word order
- syntax
- negation scope
- irony and sarcasm
- who did what to whom

Bag-of-Words Is an Abstraction

- Bag-of-words is not "stupid"
- It deliberately discards some information
- It keeps information that is often useful:
  - lexical choice
  - frequency
  - relative emphasis
  - co-occurrence across documents
- Whether this abstraction is appropriate depends on the research question

Sparse High-Dimensional Data

- Most corpora have many unique terms
- Most documents contain only a small fraction of all terms
- Result: document-term matrices are usually sparse
- This affects modeling, storage, computation, and interpretation

Multinomial Model of Language

- Text $\approx$ draw from a Multinomial distribution

Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- n = number of trials
- k = number of successes
- p = success probability

Multinomial Distribution

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}$$

where: n = text length, k = vocabulary size, p_i = probability of word i

Binomial vs. Multinomial

- **Binomial:** one word (success/failure)

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- **Multinomial:** multiple words/categories

$$P(x_1, \ldots, x_k) = \frac{n!}{x_1! \cdots x_k!} \prod_{i=1}^{k} p_i^{x_i}$$

Example Texts

- Text 1 = banana banana banana banana chocolate
- Text 2 = chocolate chocolate chocolate banana fudge
- Text 3 = banana banana
- Text 4 = ice-cream ice-cream fudge ice-cream banana chocolate
- Text 5 = fudge fudge fudge chocolate
- Text 6 = ice-cream chocolate ice-cream fudge fudge fudge

Example Texts

John
Text 1 = banana banana banana banana chocolate
Text 2 = chocolate chocolate chocolate banana fudge
Text 3 = banana banana

Mary
Text 4 = ice-cream ice-cream fudge ice-cream banana chocolate
Text 5 = fudge fudge fudge chocolate
Text 6 = ice-cream chocolate ice-cream fudge fudge fudge

Document-Term Matrix

	banana	chocolate	fudge	icecream
Doc 1	4	1	0	0
Doc 2	1	3	1	0
Doc 3	2	0	0	0
Doc 4	1	1	1	3
Doc 5	0	1	3	0
Doc 6	0	1	3	2

Document-Term Matrix

	banana	chocolate	fudge	icecream
John Rates	7	4	1	0
Mary Rates	1	3	7	5

Language Models

John Language Model

John Rates 7 4 1 0

Mary Language Model

Mary Rates 1 3 7 5

Language Models

John Language Model

John Rates	$\frac{7}{12}$	$\frac{4}{12}$	$\frac{1}{12}$	$\frac{0}{12}$

Mary Language Model

Mary Rates	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{7}{16}$	$\frac{5}{16}$

New Texts

- `new_text_1` = ice-cream fudge fudge
- `new_text_2` = chocolate chocolate banana banana

Question: Probability they were generated by John or Mary?

New Texts Document-Term Matrix

	banana	chocolate	fudge	icecream
New Text 1 Rates	0	0	2	1
New Text 2 Rates	2	2	0	0

Probability Spoken by John

- `new_text_1` = "ice-cream fudge fudge"

$$\Pr(b = 0, ch = 0, f = 2, i = 1) = \frac{3!}{0!0!2!1!} \times (0.583)^0(0.333)^0(0.08)^2(0)^1 = 0$$

Probability Spoken by John

- `new_text_2` = "chocolate chocolate banana banana"

$$\begin{aligned}\Pr(b=2, ch=2, f=0, i=0) &= \frac{4!}{2!2!0!0!}(0.583)^2(0.333)^2(0.08)^0(0)^0 \\ &= 6(0.583)^2(0.333)^2 \\ &\approx 0.23\end{aligned}$$

Probability spoken by Mary

- `new_text_1` (Mary): $\Pr = 0.18$
- `new_text_2` (Mary): $\Pr = 0.0008$

Vector Space Model

- Represent texts as vectors in multidimensional space
- Each dimension corresponds to a feature
- In a document-term matrix, features are usually words or phrases
- Once texts are vectors, we can compare them geometrically

Multidimensional Spaces

Multidimensional?

- 2D: position on a map
  X = longitude, Y = latitude
- 3D: position in the real world
  X = longitude, Y = latitude, Z = height
- 4D:
  X = longitude, Y = latitude, Z = height, T = time

Multidimensional Spaces

- The number of dimensions is the number of features needed to describe an object
- In geography: coordinates describe location
- In text analysis: words/features describe the position of a document in linguistic space

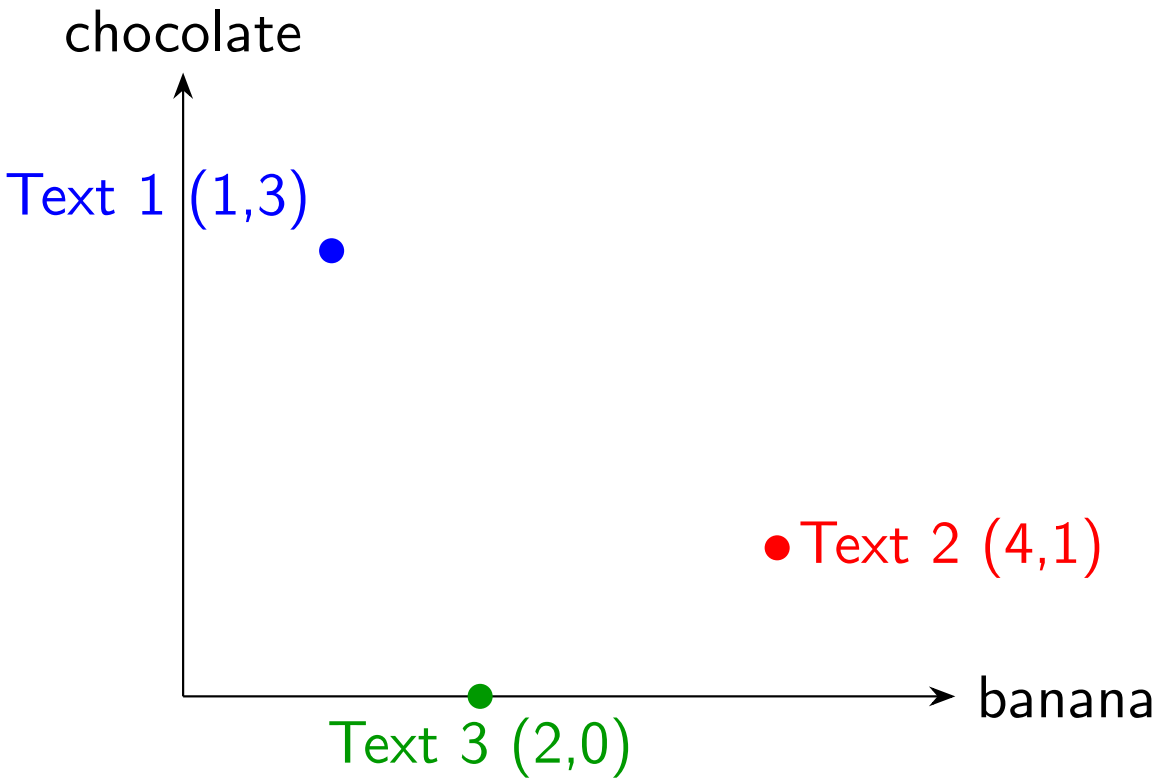
Two-Dimensional Space

- Vocabulary = {"banana", "chocolate"}
- Text 1 = "chocolate chocolate chocolate banana"
- Text 2 = "banana banana banana banana chocolate"
- Text 3 = "banana banana"

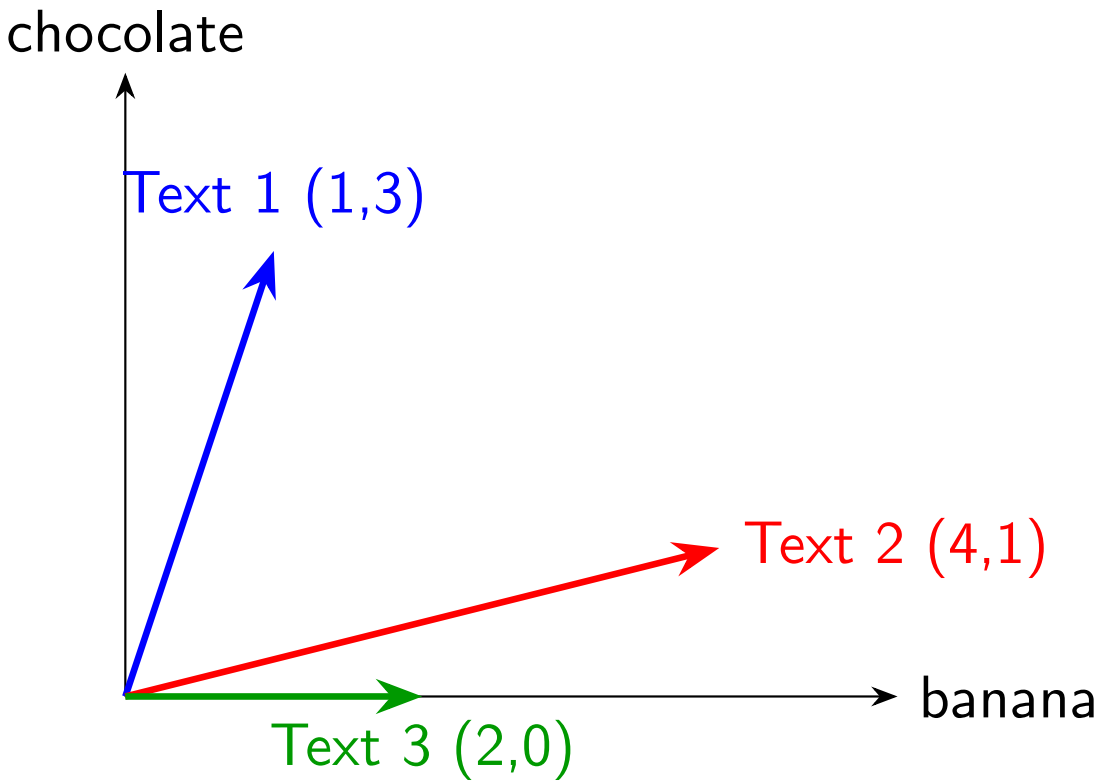
Two-Dimensional Space

- Vocabulary $= \{$"banana", "chocolate"$\}$
- Text 1 = "chocolate chocolate chocolate banana" $\Rightarrow (1, 3)$
- Text 2 = "banana banana banana banana chocolate" $\Rightarrow (4, 1)$
- Text 3 = "banana banana" $\Rightarrow (2, 0)$

Two-Dimensional Vector Space



Two-Dimensional Vector Space



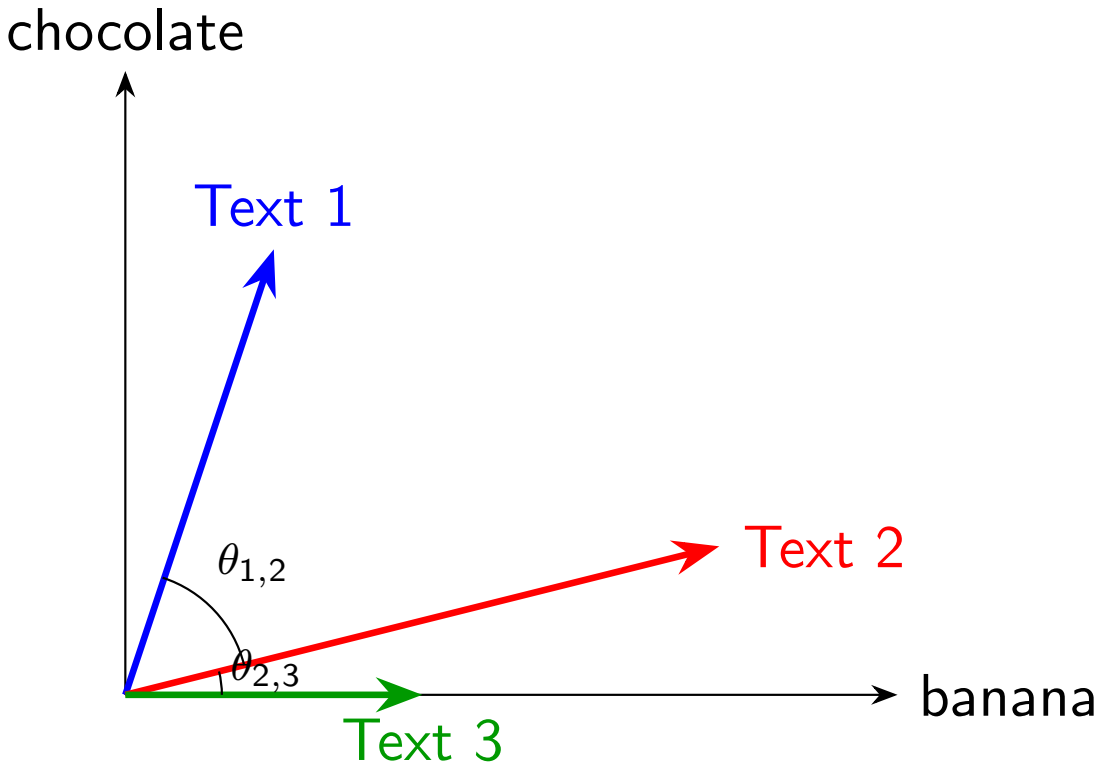
Cosine Similarity

$$\cos(\theta) = \frac{x \cdot y}{\|x\|\|y\|}$$

- Measures the angle between two vectors
- Commonly used for document similarity
- Less sensitive to document length than raw dot products

$$\cos(\theta_{1,2}) = \frac{1 \cdot 4 + 3 \cdot 1}{\sqrt{1^2 + 3^2}\sqrt{4^2 + 1^2}} = \frac{7}{\sqrt{10}\sqrt{17}} \approx 0.538$$

Two-Dimensional Vector Space with Angles



Vector Distances

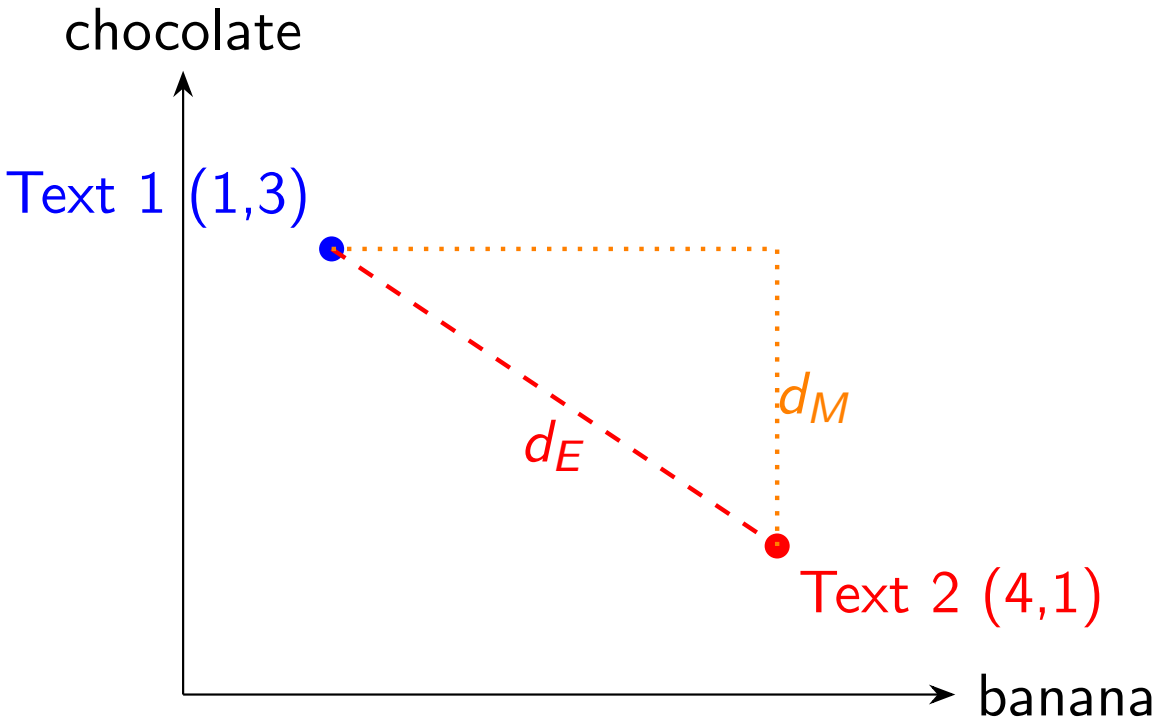
$$d_E(x, y) = \sqrt{\sum_i (x_i - y_i)^2} \qquad d_M(x, y) = \sum_i |x_i - y_i|$$

For Text 1 $(1, 3)$ and Text 2 $(4, 1)$:

$$d_E(T1, T2) = \sqrt{(1-4)^2 + (3-1)^2} = \sqrt{13} \approx 3.61$$

$$d_M(T1, T2) = |1-4| + |3-1| = 5$$

Euclidean vs. Manhattan Distance



From Two Dimensions to Thousands

- Real vocabularies are large
- A document-term matrix can easily have tens of thousands of columns
- Same logic as the two-dimensional example
- Same math, but no longer visually intuitive

Text Preprocessing

- Texts are highly dimensional
- Preprocessing reduces dimensionality and standardizes features
- But preprocessing is not neutral
- It decides which distinctions are signal and which are noise

Common Preprocessing Choices

- Punctuation
- Numbers
- Lowercasing
- Stemming / lemmatization
- Stop-word removal
- N-grams
- Removal of words by frequency

Based on common preprocessing decisions discussed in text-as-data workflows, including Denny & Spirling (2018).

Punctuation / Numbers / Lowercasing

- Often we remove punctuation and/or numbers
- Often we lowercase all text
- This reduces dimensionality
- But it can remove meaningful distinctions

Is Trump the same as trump?
Is `Article 5` meaningful?
Does `!!!` matter in social media text?
Is `COVID-19` just noise?

Lowercasing

Can help

- reduces vocabulary size
- combines accidental variants
- useful for many BoW tasks

Can hurt

- names and entities
- acronyms
- German nouns
- stylistic emphasis

Stemming / Lemmatization

- A stem is the part of the word responsible for lexical meaning
- A stem is the invariable part of the word under inflection
- "wait" is a stem of:
  - waiting
  - waits
  - waited
- A **lemma** is the base or dictionary form
  - "went", "gone" $\Rightarrow$ "go"

Stemming / Lemmatization: Why It Matters

- Useful for dimension reduction
- Often helps when measuring broad lexical content
- But it can erase meaningful stylistic or grammatical distinctions

Question

When would you want to keep inflected forms separate?

Stop Words

- Words filtered out before analysis begins
- Usually function words:
  - the
  - is
  - that
  - and
- Domain-specific words are also often excluded
  - e.g., "climate" in a corpus about climate change

Stop Words Are Not Always Useless

- Function words may be crucial for authorship attribution
- Pronouns may matter for populism, identity, or blame attribution
- Negation words matter: not, never, no
- Modal verbs may matter: must, should, can

N-grams

- So far, we have mostly used unigrams: individual words
- Texts can also be represented as sequences of *n* words
- Example: "I love ice-cream and bananas"
  - unigrams: "I", "love", "ice-cream", "and", "bananas"
  - bigrams: "I love", "love ice-cream", "ice-cream and", "and bananas"

Why N-grams?

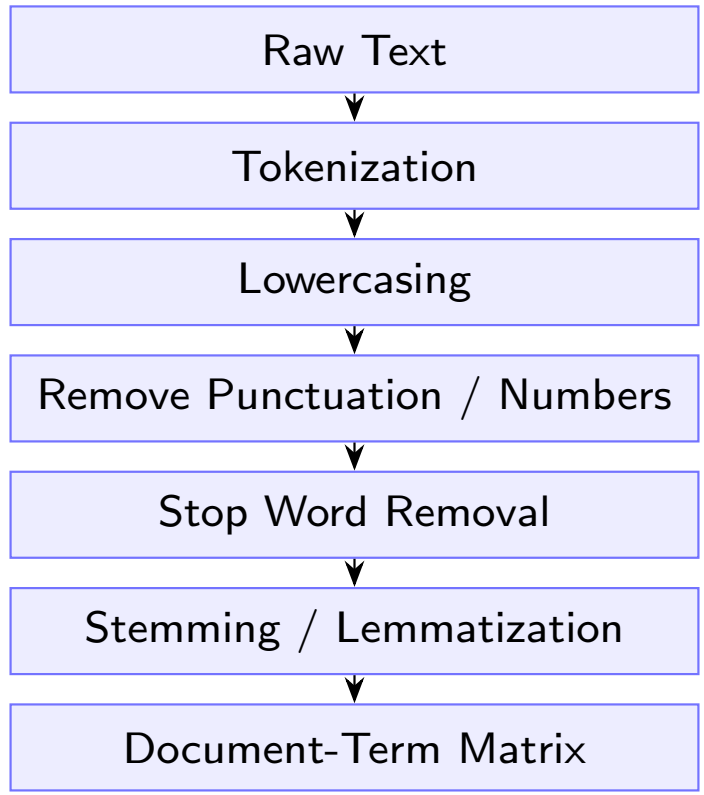
Unigrams	**N-gram meaning**
climate, change	climate change
fake, news	fake news
not, good	not good
European, Union	European Union
social, media	social media

- N-grams partly recover information lost by bag-of-words

Removal of Terms by Frequency

- Further dimensionality reduction can be achieved by removing:
  - very frequent terms
  - very infrequent terms
- Very frequent terms may not discriminate between documents
- Very infrequent terms may have low statistical power
- But rare terms can still be substantively important

Preprocessing Order Matters



Preprocessing Changes the Estimand

- Suppose we remove punctuation, numbers, stop words, and word order
- We are no longer measuring "language use" in general
- We are measuring lexical content after discarding:
  - style syntax discourse markers
  - many forms of emphasis some forms of negation
- This may be exactly what we want, but it should be a choice, not a default

Beyond Raw Counts

- A document-term matrix usually contains raw counts
- But raw counts are not always ideal
- Very frequent words dominate
- Longer documents have larger counts
- Some words are common everywhere and not very distinctive

TF-IDF

- We can do more than just count words
- We can transform counts using weights
- TF-IDF stands for:
  - term frequency
  - inverse document frequency
- Intuition: important terms are frequent in a document but rare across the corpus

TF-IDF: Term Frequency – Inverse Document Frequency

- **TF:** how often a term appears in a document
- **IDF:** downweight terms that appear in many documents

$$\text{TF-IDF}(t, d) = \underbrace{\frac{\text{count}(t, d)}{\sum_{t'} \text{count}(t', d)}}_{\text{TF}} \times \underbrace{\log \frac{N}{\text{df}(t)}}_{\text{IDF}}$$

- N = total number of documents
- $\text{df}(t)$ = number of documents containing term t

TF-IDF: What the hell?

- It looks arbitrary because it partly is

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TF-IDF: What the hell?

- It looks arbitrary because it partly is
- Introduced by Karen Spärck Jones in 1972
- Not derived from a deep theory of meaning
- It is a useful heuristic: frequent in this document, rare elsewhere
- Often works surprisingly well as a baseline

What TF-IDF Is Useful For

- Search and information retrieval
- Finding characteristic words in documents
- Document similarity
- Clustering
- Baseline features for supervised classification

What TF-IDF Is Not

- It is not a direct measure of how much a concept appears
- It downweights common words even if they are theoretically important
- It changes the interpretation of word counts
- A high TF-IDF word is distinctive, not necessarily substantively important

Comparing Texts

Once we represent texts as vectors, we can compare them.

- Which documents are most similar?
- Which documents are outliers?
- Which documents cluster together?
- Which documents are closest to a query?

Comparing Groups

Sometimes we do not only compare documents.

- Which words are distinctive of one group?
- Which words distinguish one corpus from another?
- Which words are associated with one party, outlet, period, or actor?

Examples:

- tabloids vs. broadsheets
- left parties vs. right parties
- before vs. after an event
- pro-government vs. opposition actors

Odds

- **Probability:** $p = \frac{\#\text{ successes}}{\#\text{ trials}}$
- **Odds:** ratio of probability of success to probability of failure

$$\text{Odds} = \frac{p}{1-p}$$

If $p = 0.8$, then:

$$\text{Odds} = \frac{0.8}{0.2} = 4$$

4-to-1 in favor

Log Odds

- Log odds are the natural logarithm of odds
- They map probabilities $(0, 1)$ to the full real line $(-\infty, +\infty)$

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

If $p = 0.8$, then:

$$\log\left(\frac{0.8}{0.2}\right) = \log(4) \approx 1.39$$

Comparing Two Groups

- Suppose we have two groups, for example John and Mary
- Each group has its own probability of using a word
- We compare their log odds

$$\text{Log Odds Ratio} = \log\left(\frac{p_1}{1-p_1}\right) - \log\left(\frac{p_2}{1-p_2}\right)$$

- Positive value: word is more associated with group 1
- Negative value: word is more associated with group 2

Worked Example: "banana"

- John says "banana" in 7 out of 12 words: $p_1 = 7/12 \approx 0.58$
- Mary says "banana" in 1 out of 16 words: $p_2 = 1/16 = 0.0625$

$$\begin{aligned}\text{Log Odds Ratio} &= \log\left(\frac{0.58}{0.42}\right) - \log\left(\frac{0.0625}{0.9375}\right) \\ &= \log(1.38) - \log(0.0667) \\ &\approx 0.32 - (-2.71) \approx 3.03\end{aligned}$$

Positive log-odds ratio $\Rightarrow$ "banana" is more associated with John.

Why Use Log Odds?

- Probabilities are bounded between 0 and 1
- Odds ratios can be very skewed
- Log odds:
  - are symmetric around 0
  - have direction and magnitude
  - are related to logistic regression
  - are useful for comparing word usage between groups
- In text analysis, log-odds approaches are often used to find distinctive words

Representation Choice Matrix

Goal	**Useful representation**	**Risk**
Measure word prevalence	raw counts / proportions	common words dominate
Find distinctive terms	TF-IDF / log odds	distinctiveness $\neq$ importance
Compare documents	TF-IDF + cosine similarity	length and preprocessing matter
Capture phrases	n-grams	feature explosion
Reduce dimensionality	trimming / stemming	loss of meaning
Measure style	keep function words / punctuation	noisy features

Representation Choices Are Research Choices

- If you remove stop words, you probably cannot study style well
- If you remove numbers, you may lose dates, prices, laws, or percentages
- If you use unigrams, you may miss fixed phrases
- If you use TF-IDF, you are measuring distinctiveness, not raw prevalence
- If you aggregate to documents, you may lose sentence-level meaning

No Universal "Correct" Preprocessing

Bad question

What is the correct preprocessing pipeline?

Better question

What preprocessing pipeline is appropriate for this research question, corpus, and validation strategy?

Beyond Bag-of-Words

- Bag-of-words represents documents by explicit word counts.
- This is simple, transparent, and often useful.
- But it treats words as independent features.
- It does not know that:
  - "car" and "automobile" are similar
  - "good" and "not good" are different
  - the same word can mean different things in different contexts
- Later in the course, we will move from count-based representations to learned representations.

Distributional Hypothesis

- **Key idea:** words that occur in similar contexts tend to have similar meanings.
- "You shall know a word by the company it keeps."
- This motivates representing words by their contexts.
- Instead of asking only how often a word appears, we ask:
  - Which words appear near it?
  - Which documents does it appear in?
  - Which contexts is it associated with?

Preview: Word Embeddings

- Word embeddings represent words as dense vectors.
- Similar words should be close together in vector space.
- Unlike bag-of-words, the dimensions are learned and not directly interpretable.
- Embeddings are useful for:
  - semantic similarity
  - measuring concepts with word lists
  - document representations
  - transformers and LLMs
- We return to this in detail on Day 4.

Practical Session: Representation Choices

1. Load a small text corpus
2. Tokenize the documents
3. Build a document-term matrix
4. Compare raw counts and TF-IDF
5. Apply different preprocessing choices
6. Compute document similarity
7. Discuss: what changed, and why?

Practical: Questions to Answer

- What are the most frequent words?
- What are the most distinctive words?
- Which documents are most similar?
- What changes after stop-word removal?
- What changes after stemming or lemmatization?
- What changes after using TF-IDF?

Practical: Compare Representations

Version	Representation
1	raw counts
2	lowercased counts
3	stop words removed
4	stemmed / lemmatized
5	bigrams included
6	TF-IDF weighted

- Same corpus
- Different representation
- Potentially different conclusion

Discussion

- Which terms disappeared after preprocessing?
- Which terms became more important after TF-IDF?
- Which documents became more or less similar?
- Which representation best fits your own research question?

Key Takeaways

- Text analysis begins with representation
- A document-term matrix is simple but powerful
- Bag-of-words is not stupid: it is a deliberate abstraction
- Preprocessing is not cleaning: it is measurement design
- TF-IDF and log odds help identify distinctive language
- Different representations support different claims

The Main Lesson

There is no raw text analysis.

There is always a representation.

A representation decides what counts as signal, what counts as noise, and what kind of claim the analysis can support.

Tomorrow

- Dictionaries and regular expressions
- Human annotation and codebooks
- Supervised machine learning
- Evaluating classification models